Curtain Call, Cab Call

Broadway Attendance and Taxi Demand in the NYC Theater District

Jiawei Cai (Ryan)
Github repo with commit

October 4, 2026

1 Introduction

Every single night, tens of thousands of theatergoers rush out of Broadway's theaters, creating one of the most predictable demand shocks in New York City's transportation network. Yet it is unclear whether that demand actually reaches taxi and ride-share drivers or is made up of walking, the subway, and proximity to hotels. This report investigates whether or not Broadway show attendance is associated with taxi (Yellow) and High Volume For-Hire Vehicle (FHVHV, including Uber, Lyft, Via and Juno) pickup demand in the Theater District, and whether or not that relationship is strong enough to be useful to a taxi driver deciding when and where to work.

1.1 Datasets

We used the 2023 and 2024 Yellow Taxi and High Volume For-Hire Vehicle trip records provided by the NYC Taxi and Limousine Commission (TLC) [1]. Green Taxis were excluded because they can only pick up passengers above West 110th Street/East 96th Street in Manhattan, which is outside of the Theater District [2]. In addition, we also excluded the data of Regular (non-high-volume) For-Hire Vehicles, because they mainly consist of pre-booked Community Car, Black Car and Luxury Limousine data, instead of the on-demand, street-hail style of transport we want to investigate [3].

We chose the years 2023 and 2024 for three reasons: (1) This range avoids the recovery years of the pandemic (2020–2022). (2) They cover two complete Broadway seasonal cycles, including the December holiday peak in each year. (3) It avoids Manhattan's congestion pricing policy that started in January 2025, which would interfere with trip volumes and fares in the zones this report analyzes.

The external datasets include the weekly gross and attendance data for all Broadway shows from the Internet Broadway Database obtained from Kaggle, as well as the data of the 41 theaters of Broadway and their street addresses [4]. These theater addresses were geocoded and joined to TLC taxi zones and gave four zones that contain a Broadway theater: Times Sq/Theatre District, Clinton East, Garment District, and Midtown Center (TLC zone IDs are 230, 48, 100, and 142).

The two trip-record datasets total over 550 million raw rows across the two years (79,479,946 Yellow Taxi and 471,960,468 FHVHV, before any filtering).

1.2 Research Goals

The target audience for this report is Yellow Taxi and FHVHV drivers deciding when and where to position themselves for a shift, and by extension, the TLC dispatch bases making the same allocation decision on a larger scale. The research goal is to know (a) whether Broadway attendance predicts taxi/FHVHV demand in the Theater District more than what a control zone would predict, (b) whether that relationship is caused by broad seasonal trends, short-term week-to-week movement, or both, and (c) whether the relationship differs between Yellow Taxis and FHVHVs.

To achieve this, we filtered the trip datasets to the four theater zones and a non-theater control zone, aggregated trips to weekly totals, and joined this with the weekly Broadway attendance values. We also tested the strength of this relationship using Pearson and Spearman correlation, isolated the relationship after removing shared seasonal trends, compared Yellow Taxi and FHVHV relationships using a Fisher z-test, and examined within-week timing (hour of day, day of week) directly. We then fit a Linear Regression and a Random Forest model to try and predict the relationship.

2 Preprocessing

2.1 Broadway theater data

- Theater addresses were geocoded via the US Census geocoding API. 59/60 matched successfully (one was a blank row). After removing duplicate rows caused by theater name changes over time and dropping all NULL/missing values, 41 theaters remained, matching the number of current theaters on Broadway.
- Geocoded theater points were spatially joined to the TLC taxi zone shapefile. Unmatched theaters were matched to the nearest zone instead of being discarded. All 41 theaters matched to a zone, yielding the four theater zone IDs used throughout this report.

2.2 Taxi trip data

Yellow Taxi and FHVHV parquet files for all 24 months of 2023–2024 were read, corrected into a consistent schema, and filtered to trips with a pickup in one of the four theater zones or in the control zone. Trips were also restricted to pickup dates from 2023-01-01 to 2024-12-31, and no NULLs were found after filtering. Several feature-engineering and filtering considerations were also made, including:

- **Unfiltered features:** The VendorID, RatecodeID and payment_type and similar features are intentionally not filtered as the analysis never uses them and the passenger_count feature was also not filtered as this is self-reported [2]. We chose not to delete them because we do not want to lose important data just because of data corruption or human error; what truly shows that a ride has been completed is the trip_distance, the feature-engineered trip_duration_secs and the financial features (fare_amount, tolls_amount, tip_amount, etc.).
- **Trip distance:** The trip_distance and trip_miles features were constrained to a minimum of 0.2 miles (the 2% percentile of all records), and a maximum of 35 miles (the longest distance between boundaries of the five boroughs [5]).
- **Trip duration:** For the Yellow Taxi data, the trip_duration feature was engineered by taking the difference in the tpep_pickup_datetime and tpep_dropoff_datetime features. The trip_duration and trip_time features are constrained from 60 to 7200 seconds (2 hours).
- **Financial features:** The financial features were all constrained to be greater than 0, and a minimum fare of 3 USD was constrained for Yellow Taxi fare amount as stated by the TLC [6].

- Raw row counts before and after filtering were also logged and summarized in Table 1.

Filter stage	Yellow Taxi rows	FHVHV rows
Before filtering	79479946	471960468
After data-constraint filtering	76110915	469327480
After zone and time filtering	8228548	16632799
Percentage left	10.35 %	3.52 %

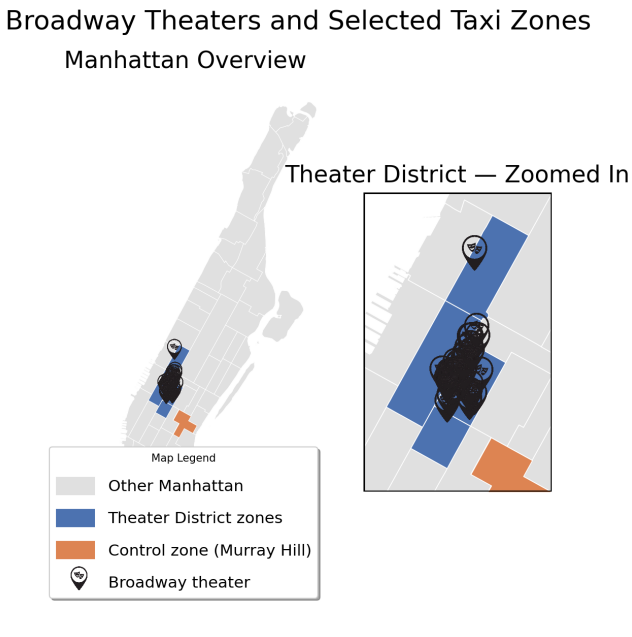
Table 1: Number of rows in each dataset before and after filtering.

- **Week ending feature:** Pickup timestamps were converted to a `week_ending` field (the Sunday that closes each Monday–Sunday week), so trips could be joined with the weekly Broadway data.
- **Final aggregation:** Trip data was aggregated to one row per week per zone group (theater zones, control zone), giving weekly trip counts, fares, and tips, with 104 rows per zone group.

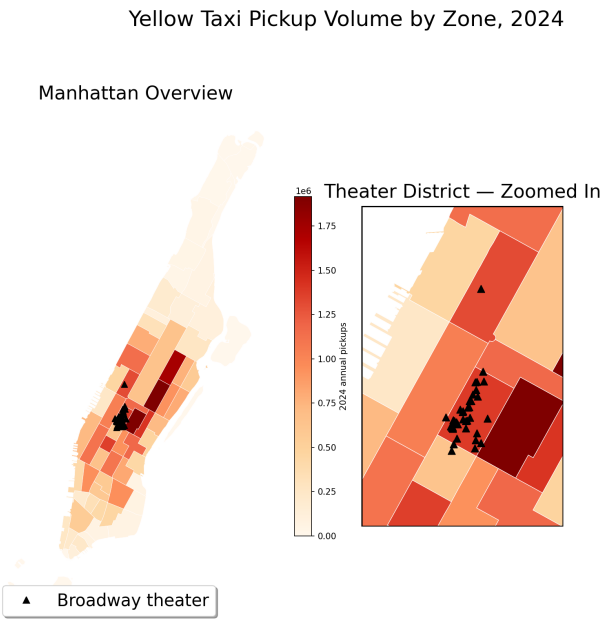
2.3 Broadway weekly grosses data

- Weekly grosses had 2259 rows and were parsed from string format into numeric `gross` and `attendance` columns, and filtered to the 104 weeks of 2023–2024.
- The filtered data is then joined with the filtered TLC data.

3 Analysis and Geospatial Visualisation



(a) Theater District, control zone, and theaters.



(b) Annual 2024 pickup volume, all Manhattan zones.

Figure 1: (a) The four Theater District zones and Murray Hill control zone with the 41 geocoded theaters shown as points. (b) How those zones compare to citywide Yellow Taxi pickup density.

Figure 1a confirms the geocoded theaters fall inside the zones this report treats as the Theater District and Figure 1b places them in citywide context.

3.1 Distribution and outliers

Weekly Theater District trip counts had a mean of 78,899 and standard deviation of 9,392 with a skew of -0.983 across all 104 weeks. An IQR plot was used to check outliers (bounds [58,156, 101,237]) and two weeks were flagged for being too low: 38,333 trips in the week ending 24 September 2023, and 57,271 trips in the week ending 31 December 2023. The September 2023 low is when the United Nations General Assembly's High-Level Week (18–26 September 2023) happened; the NYC Department of Transportation recorded the slowest Midtown traffic speeds of the year during this week (under 4 mph) because of street closures and motorcade security [7]; Broadway attendance that week was 198,785 (comparable to nearby weeks), so the drop is likely caused by the street closures and not because of Broadway attendance. The December 2023 low is the first of two Christmas-weeks found in this data; this is discussed together with the 2024 Christmas week in Section 3.9.

3.2 Relationship with Broadway attendance

As shown in Table 2, weekly Yellow Taxi trips in the Theater District are correlated with weekly Broadway attendance. Weekly fares correlated more strongly than raw trip counts, while weekly tips correlated somewhat less strongly, suggesting the relationship holds for trip value and trip volume.

	Pearson	Spearman
Weekly trips	$r = 0.436$, $p < 0.0001$	$r = 0.311$, $p < 0.0001$
Weekly fares	$r = 0.482$, $p < 0.0001$	$r = 0.479$, $p < 0.0001$
Weekly tips	$r = 0.381$, $p < 0.0001$	$r = 0.390$, $p < 0.0001$

Table 2: Pearson and Spearman correlation values of Yellow taxi data and Broadway data

3.3 Is this Broadway-specific, or general Midtown seasonality?

A comparable non-theater Manhattan zone (Murray Hill) also correlated with attendance, but more weakly ($r = 0.311$, $p = 0.0013$) (a difference of 0.125 against Theater District). This shows that the Theater District relationship is partly attributed to the general seasonal trend that affects both zones.

3.4 Trend versus week-to-week movement

When the shared seasonal trend was removed by correlating week-over-week percentage changes instead of the raw values, the relationship was not significant ($r = -0.025$, $p = 0.803$), showing that the strong correlation in the raw data is partly because both series follow the same yearly pattern. It does not necessarily mean that a busy week on Broadway leads to more taxi pickups.

3.5 Yellow Taxi versus FHVHV

FHVHV pickups in the same zones showed no meaningful correlation with attendance ($r = 0.013$, $p = 0.899$; Spearman $\rho = 0.002$, $p = 0.986$). A Fisher z-test comparing the Yellow Taxi and FHVHV correlation coefficients found them significantly different ($z = 3.229$, $p = 0.0012$).

To check whether the FHVHV result is Broadway-specific, we extended the control-zone comparison to FHVHV as well, which is summarized in Table 3.

A Fisher z-test between FHVHV's two zones found no significant difference ($z = 1.129$, $p = 0.259$). Combined with Table 3, FHVHV's weak correlation in the Theater District cannot be distinguished from its correlation in a zone with no Broadway theaters at all. Unlike Yellow Taxi, FHVHV demand shows no detectable Broadway-specific signal at either zone.

	Theater District	Murray Hill (control)
Yellow Taxi	$r = 0.436$, $p < 0.0001$	$r = 0.311$, $p = 0.0013$
FHVHV	$r = 0.013$, $p = 0.899$	$r = -0.145$, $p = 0.141$

Table 3: Correlation with weekly Broadway attendance, by taxi type and zone.

3.6 Does theater density within the aggregate matter?

The four theater zones do not have the same amount of theaters (37 fall in Times Sq/Theatre District alone, with only 2, 1, and 1 in the other three). Since three of the four zones are mainly non-theater Manhattan activity that incidentally contains one or two theaters, aggregating all four together will dilute any Broadway-specific signal towards general noise. We tested this by rebuilding the weekly series using only the Times Sq/Theatre District and comparing it against the full aggregate. The core zone alone correlated with attendance at $r = 0.445$ ($p < 0.0001$), which is not significantly different from the full aggregate's $r = 0.436$ (Fisher $z = 0.085$, $p = 0.932$). Despite the large differences in the number of theaters across zones, both the core zones and the full set of zones show the same overall pattern. This consistency supports the use of the theater zone definition throughout the report.

3.7 Timing within the day

Figure 2 plots Theater District pickups by hour of the day. Pickups are highest at 10pm, with 614,943 trips across the two years, accounting for 7.5% of the daily total. This lines up with the end of a typical evening Broadway show, which usually runs from around 7–8pm until 9:30–10:30pm. This is different from the control zone which peaks four hours earlier at 6pm and then declines steadily. The Theater District has 20.0% of its daily pickups from 9pm-midnight, versus only 12.9% for the control zone.

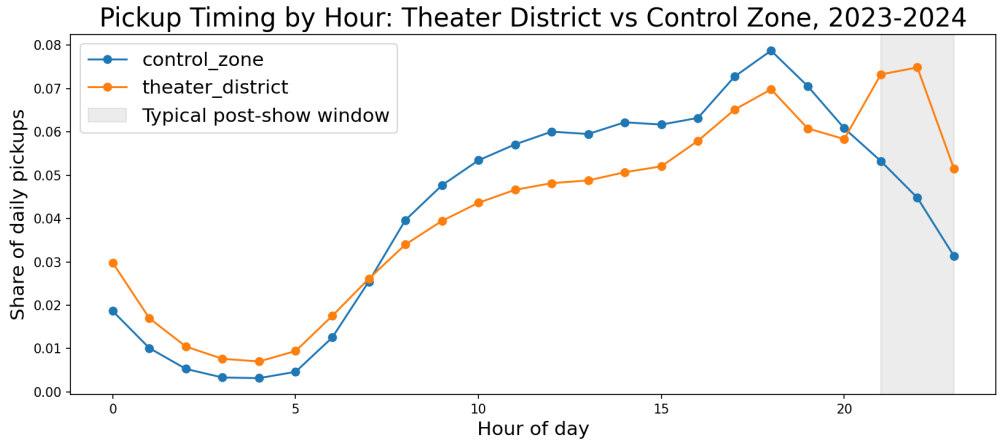


Figure 2: Pickup Timing by Hour: Theater District vs Control Zone, 2023-2024.

3.8 Timing within the week

Thursday (not a matinee day) was the busiest day overall (1,288,508 trips), slightly ahead of Wednesday and Saturday (1,280,786 and 1,237,857), both matinee days. Sunday, also a matinee day, was the second lowest (1,048,109), ahead of only Monday (977,274). Any matinee effect is therefore secondary to a broader weekly pattern.

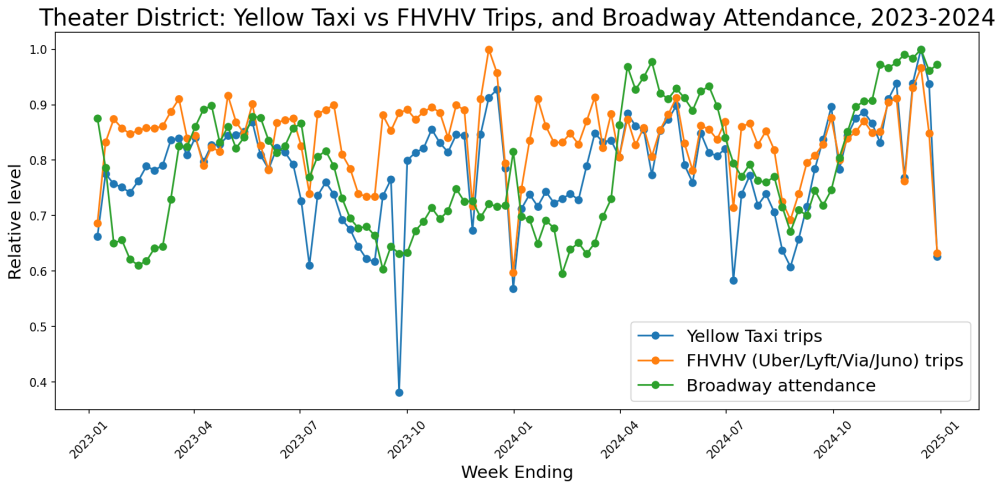


Figure 3: Theater District: Yellow Taxi vs FHVHV Trips, and Broadway Attendance, 2023-2024

3.9 Case study: the final week of the year

As seen in Figure 3, the week ending December 29, 2024 saw a sharp drop in Theater District taxi trips, falling from 94,497 the previous week to 63,124 (-33.2%), even though Broadway attendance remained near its yearly peak at 306,180. A very similar pattern occurred the year before when trips fell from 79,124 to 57,271 (-27.6%) in the week ending December 31, 2023 while Broadway attendance actually increased to 256,751. The same pattern in both years suggests that this is a recurring seasonal pattern. This also matches published findings that Christmas is typically one of the quietest weeks of the year for NYC taxis [8].

The control zone saw a larger drop than the Theater District in both years, although the difference was much bigger in 2024 (-44.5% in 2024, -29.9% in 2023). The pattern is consistent across both years, suggesting that Broadway-related demand may offset the broader holiday slowdown. However, the size of this difference varied between years, so 2024 should not be treated as the typical effect.

4 Modelling

We fit two contrasting models to predict weekly Theater District Yellow Taxi trips from Broadway attendance, gross, and calendar features (month, week of year and a holiday-season flag for Nov/Dec):

$$\text{weekly_trips} = f(\text{attendance}, \text{gross}, \text{month}, \text{week_of_year}, \text{is_holiday_season}) + \epsilon \tag{1}$$

Linear Regression represents a simple baseline, while Random Forest (200 trees, max depth 4, chosen to limit overfitting given only 52 training weeks) allows non-linear interactions. Models were trained on all 52 weeks of 2023 and tested on all 52 weeks of 2024, plotted in Figure 4. Month and week-of-year were encoded as raw integers rather than cyclically, as the training data only consists of 1 year.

Model	MAE	RMSE	R^2
Linear Regression	5,885	7,166	0.369
Random Forest	6,517	8,784	0.051

Table 4: Test-set performance for both models, trained on 2023, tested on 2024

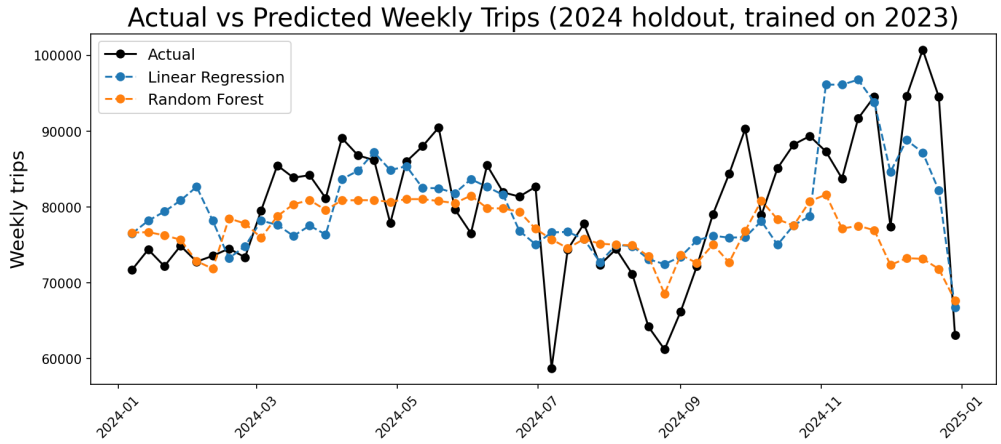


Figure 4: Actual versus predicted weekly Theater District trips, 2024 holdout, trained on 2023.

From Table 4, Linear Regression explains 36.9% of the variance on the 2024 data, compared with just 5.1% for Random Forest. The higher error from Random Forest is likely because tree-based models are limited to the range of values they trained on and cannot extrapolate beyond it. In this case, the 2024 peak week reached 100,737 trips, which was higher than every week in the 2023 training data. Linear Regression can continue its trend beyond the training range, giving it an advantage for dispatchers forecasting December, where Random Forest's ceiling effect would cause under-supply in exactly the weeks drivers stand to earn the most.

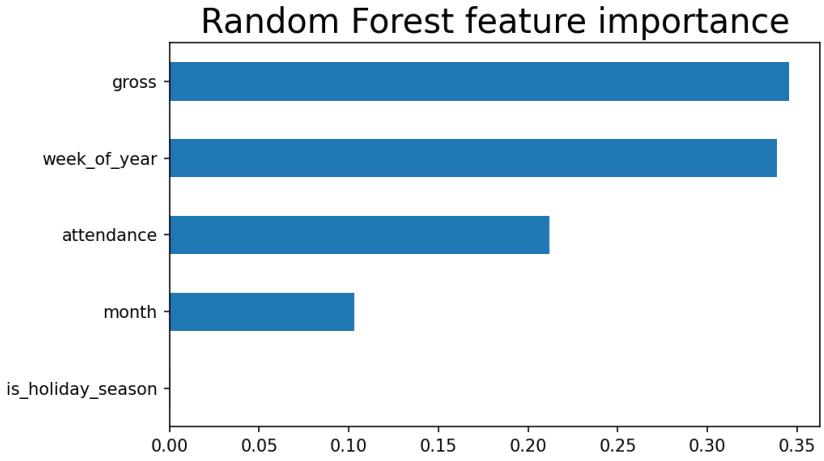


Figure 5: Random Forest feature importance for predicting weekly Theater District trips.

Figure 5 shows which features the Random Forest depends on the most, with `week_of_year` dominating, suggesting that the model is mostly learning the seasonal trend rather than attendance. This is consistent with the de-trended correlation being non-significant.

5 Discussion

Overall, the results suggest that Broadway attendance is more closely linked to taxi demand in the Theater District than in a similar zone without theaters. The relationship is also stronger for Yellow Taxis than for FHVHV trips. However, once we remove the overall seasonal trend, the connection

between Broadway attendance and taxi demand becomes much weaker, as both tend to increase as the holiday season approaches and decrease afterward, but a particularly busy Broadway week does not necessarily mean that there will be more taxi pickups that same week. The correlation is likely influenced by a similar seasonal pattern. Additionally, the Yellow Taxi/FHVHV gap is likely because Broadway's audience skews towards a demographic more likely to street-hail a cab than to order an app-based ride. Another reason is that FHVHV demand is influenced by a mix of the seasonal trend and generic Midtown activity that dilutes any Broadway-specific signal. Distinguishing between these would require trip-level data on rider characteristics that TLC does not publish, and is a reasonable direction for further work. The modelling results support the same picture. Linear Regression explained 36.9% of 2024's weekly variation. This shows that there is a meaningful pattern that can be learned from previous seasonal data. Random Forest explained only 5.1%. This is because tree-based models struggle when the new data goes beyond the range they saw during training. This highlights the importance of choosing a model that suits the data and the forecasting task.

6 Recommendations

1. Use Broadway as a seasonal and intra-week timing signal. The de-trended relationship is not significant ($r = -0.025$, $p = 0.803$), confirming that drivers should not expect a single strong week of Broadway sales to mean that there will be more pickups that week. Positioning decisions should be based on month-to-month seasonal patterns (e.g. high demand from autumn to early December). However, within a given day, the Theater District has 20.0% of its daily pickups in 9pm to midnight window (versus 12.9% for a comparable zone). Treating all Midtown evenings the same misses the clear spike in demand around Broadway curtain times, which does not appear in other zones.

2. Expect Christmas week itself to be an exception to the usual holiday pattern. This happened in both 2023 and 2024: taxi trips dropped sharply during the week of Christmas, even though Broadway attendance remained high or even increased. This goes against our initial assumption that busier Broadway weeks always mean more taxi demand. Drivers should treat Christmas week as an off-peak week rather than a holiday-season peak, maybe by deprioritizing Theater District positioning in favour of routes tied to non-Broadway holiday travel (airports, hotels) and any demand-forecasting tool should flag Christmas week explicitly rather than extrapolating through it.

7 Conclusion

Overall, Broadway attendance has a stronger relationship with taxi demand in the Theater District than in the comparable control zone ($r = 0.436$ vs. $r = 0.311$). The relationship is also much stronger for Yellow Taxis than for FHVHV ($r = 0.436$ vs. $r = 0.013$), confirmed by the Fisher z-test. We also checked whether having most theaters concentrated in one zone could distort the results, but found no evidence that it did. The main finding is that this relationship is mostly seasonal rather than week-to-week. Once the shared seasonal trend is removed, Broadway attendance no longer predicts taxi demand in a given week. The timing within each day is clearer as Theater District pickups peak around four hours later than in the control zone, lining up with the time Broadway shows finish. The modelling results also show that the seasonal pattern can be predicted reasonably well when the model has seen a previous holiday season, with Linear Regression explaining 36.9% of the variation in 2024. Random Forest performed much worse because it struggles when demand goes beyond the range seen during training. For drivers and dispatchers, the takeaway is that Broadway attendance is more useful as a seasonal indicator than as a real-time signal, and the busiest period to focus on is roughly 9pm–midnight in theater zones. Christmas week should also be treated separately, and forecasting models should be able to extrapolate beyond its training range.

References

[1] NYC Taxi and Limousine Commission. *TLC Trip Record Data*. https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page. Accessed: 2026-08-22.

[2] NYC Taxi and Limousine Commission. *TLC Trip Records User Guide*. https://www1.nyc.gov/assets/tlc/downloads/pdf/trip_record_user_guide.pdf. Accessed: 2026-08-22.

[3] NYC Taxi and Limousine Commission. *For-Hire Vehicle Bases*. https://www.nyc.gov/site/tlc/businesses/for-hire-vehicle-bases.page. Accessed: 2026-08-29.

[4] Jesse Mostipak and The Broadway League. *Broadway Weekly Grosses*. https://www.kaggle.com/datasets/maggieakarn/broadway-datasets. Accessed: 2026-08-21.

[5] Fiona F. *NYC by the Numbers*. https://www.walksdevour.com/blog/nyc-by-the-numbers/. Accessed: 2026-08-30.

[6] NYC Taxi and Limousine Commission. *Taxi Fare*. https://www.nyc.gov/site/tlc/passengers/taxi-fare.page. Accessed: 2026-08-29.

[7] NBC New York. *UN General Assembly Gridlock Alert: Street closures, traffic changes and what to know*. https://www.nbcnewyork.com/news/local/un-general-assembly-gridlock-alert-street-closures-traffic-changes-and-what-to-know/4680265/. Accessed: 2026-08-29.

[8] Gaurav Singh. *Forecasting NYC Yellow Taxi Ridership Decline: A Time Series Analysis of Daily Passenger Counts (2017-2019)*. https://arxiv.org/html/2507.10588. Accessed: 2026-08-25.